

Experiments (Review)

STAT 24510-30040

(Winter 2025)

In class, we used some technical terms, such as treatments, conventionally used in experimental design studies.

In the following we review basic concepts in the study of design of experiments, which occur regularly in basic assumptions in two sample tests, analysis of variance, and linear regression models.

Experimental design

An experiment is a study in which an “experimenter” deliberately **assigns treatments** to units.

In a **randomized** design, a probabilistic rule is used to assign treatments to units.

Randomization tries to assure that the units getting different treatments do not systematically differ.

Observational study

Observational studies are the ones where researchers draw inferences from a sample to a population where the independent variable is not under the control of the researchers, usually because of ethical concerns or logistical constraints.

In an observational study, the researchers **observe** the effect of a risk factor, diagnostic test, treatment or other intervention without trying to change who is or isn't exposed to it.

Examples

- Case control
- Cohort studies

Randomization

In design of experiments, **randomization** refers to random allocation of the experimental units across the treatment groups, that is, random assignments of objects or participants to the treatment in an experiment.

- How is it done?
Study participants or subjects are assigned to a treatment group by probability alone.
- What purpose does it serve?
Randomization minimizes the differences among groups by equally distributing subjects with particular characteristics among sample groups.
- Where is it used?
 - Sample surveys
 - Experimental designs

Completely randomized design (CRD)

An experiment has a completely randomized design if a predetermined number of replication of each treatment are assigned to experimental units completely randomly, that is, each allocation of treatments to units is equally likely.

- For example, if a coin toss is used to decide between treatment and control for 20 patients, this is not a CRD, as the number of replications in the control is random.
- If the patients draw lots, say, draw from 20 tickets in a hat, 10 of which are marked “treatment”, it is a CRD.

Experimental units vs Measurement units

Experimental units are the smallest division of the experimental material that treatments are assigned.

Any two experimental units should be able to be assigned to different treatments.

Measurement Units are where the measurements of the response variable are actually made on.

Often the measurement units are the same as the experimental units.

Sometimes a measurement unit is a subset of an experimental unit.

Example (Experimental units vs Measurement units)

Two teaching methods A and B (treatments) are randomly assigned to different classes.

The comparison of the teaching methods will be based on test scores of the students.

In this study, the students are the measurement units, while the classes are the experimental units.

Control

In many experiments, one treatment might be a control level that should not affect response of interest.

To reduce possibilities for misleading results from medical experiments on humans, the design often use

- Placebo control (e.g., a pill with no active ingredient)
- Double-blind treatment assignment in which neither patient nor doctor knows whether patient received active treatment or control.

Correlation vs Causation

In observational studies, where researchers simply collect existing data and infer based on the data collected, there is always the worry that the subjects getting different treatments systematically differ already, leading to erroneous conclusions about impacts of drugs or procedures.

In practice, when designed experiment is not possible, researchers seek the situation that mimic an experiment, such as when the treatment and control conditions naturally occurred, not designed or controlled by the investigators, however the process governing the treatment and control assignments arguably resembles that of a random assignment.

Related sections in Rice's text: 11.4, 12.3.